

Augmented Study Room

Project Report

Augmented Reality – 108871

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1 Introduction

Augmented Study Room is a mobile augmented reality application that associates digital content with recognizable elements in a study room.

The project combines two augmentation modes: an informational component, implemented through textual interfaces and a spatial component, consisting of 3D models placed on the floor or on the surface associated with the target.

2 Libraries and SDKs

The main components used in the project are:

- **AR Session and XR Origin:** manage the session lifecycle, the reference coordinate system, and the camera pose within the environment reconstructed by the device;
- **AR Tracked Image Manager:** connects the reference image library to the image-tracking subsystem and provides each target's identifier, pose, physical dimensions, and tracking state;
- **AR Plane Manager:** initially detects planes in the environment, which are used to estimate the floor height;
- **ARKit XR Plugin:** translate AR Foundation operations into the native features of iOS;
- **Unity Input System:** manages events from the touchscreen and pointer;
- **Unity UI and TextMesh Pro:** implement the information cards, navigation controls, and text rendering.

3 Augmented Reality Flow

3.1 Content Recognition and Activation

The application contains five reference images, each associated with its real-world physical dimensions. This metadata allows the subsystem to determine the target's metric scale and estimate its pose. The configuration supports up to five simultaneously tracked moving images.

At runtime, ARKit analyzes the camera frames and compares them with the descriptors in the library. When a match is detected, AR Foundation returns a trackable object characterized by a persistent identifier, its pose in the XR Origin coordinate system, its estimated dimensions, and a tracking state. The application uses the reference name to select the information card and any 3D model that must be instantiated.

Content is visible only while the target is in the *Tracking* state. If pose estimation becomes limited or the target leaves the camera's field of view, the instances are disabled but not destroyed. When the target is recognized again, the card's local state is preserved.

3.2 Information Card Placement

The target pose is evaluated to distinguish between horizontal and vertical images. The criterion compares the target's local normal with the world's vertical axis; beyond a threshold of 45°, the image is classified as vertical. For a horizontal target, the card is placed above the reference; for a vertical target, it is shifted to the side. In both cases, the marker remains visible and selectable.

The card's transform is updated every frame based on the image's current pose. A billboard behavior orients its surface toward the camera, using the camera's vertical axis as a reference to reduce awkward and hard-to-read tilting. The card therefore remains anchored to the target in position while being oriented toward the viewer to improve readability and interaction.

3.3 3D Content Placement

Unlike the information cards, the 3D models are not directly anchored to the target pose and must appear stable within the environment. Two placement modes have been defined:

- **OnFloor:** the height is obtained from the lowest detected plane so that the object can be placed on the floor;
- **OnSurface:** the model retains the target's height, making it suitable for a supporting surface such as a desk or shelf.

In these modes, position and orientation are fixed after the first valid placement. This prevents small fluctuations in the image's estimated pose from being transferred to the model. Plane detection is disabled after five seconds so that newly detected surfaces cannot move the model.

3.4 User Interaction

Each card contains three pages, only one of which is active at a time. The previous and next controls update the current page, while three indicators provide visual feedback on the user's position within the card.

The close command disables both the card and its associated model and records the choice for the specific tracked target. The content can be reopened by tapping the physical target: a ray is generated from the selected screen point and intersected with the image's mathematical plane. The intersection point is transformed into the target's local coordinates and compared with its dimension. If the target is lost and subsequently recognized again, the closed state is reset.

3.5 Scripts Involved

The application's behavior is managed by three scripts, each dedicated to a specific aspect.

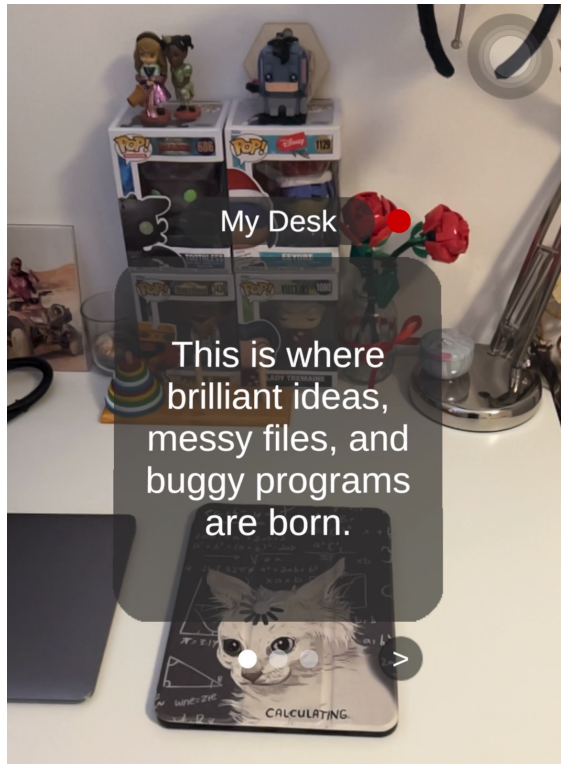
- **TrackedImageContentController.cs:** checks which images are recognized by the camera and displays their associated content. It places the information cards according to the orientation of each reference and positions any 3D models on the floor or on a surface. It also hides content when the reference is no longer visible and displays it again when the image is framed or tapped by the user.
- **CardPager.cs:** manages navigation within the information cards. It allows users to move between pages using the previous and next buttons while displaying only one page at a time. It also updates the arrows and graphical indicators to clearly communicate the current position within the card.
- **CardHider.cs:** connects the close button on the card to the main system. When the button is pressed, it identifies the selected card and requests that both the card and any associated 3D model be hidden. The content can be reopened by tapping the physical image again.

4 Augmented Content

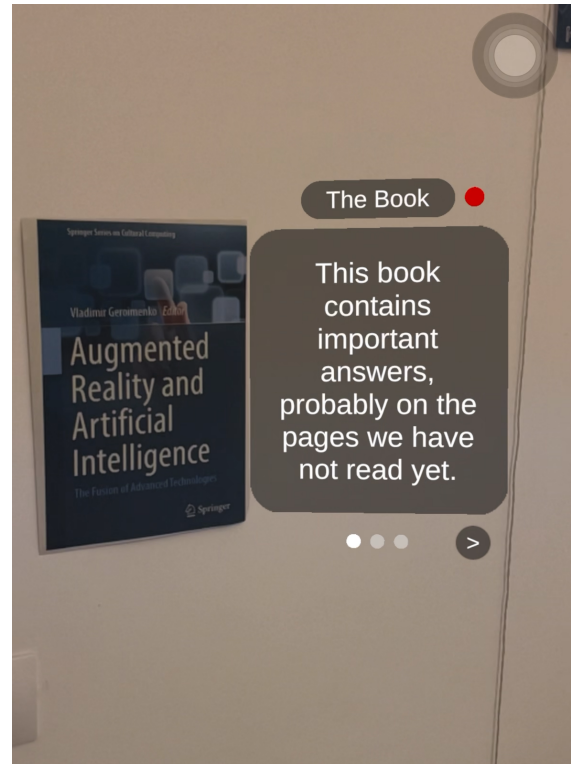
Five images associated with augmented reality content have been placed in the room. Each target image activates an information card and, where applicable, a dedicated 3D model. The augmented contents are as follows:

- **Desk:** presents the student's work and study area. The recognized image is the cover of a tablet case.
- **Crying Corner:** offers a short, humorous description of the challenges of university life. Together with the card, a 3D model of a weeping angel is displayed on the detected floor.
- **Book:** displays a card dedicated to a book about augmented reality and AI.

- **Lewis Hamilton:** summarizes the final events of the 2021 Formula One World Championship and the controversies surrounding them. A 3D trophy is placed on the target's surface to symbolically represent the world title Hamilton (never) won.
- **Tiziano Ferro:** contains a card dedicated to the artist and the memory of the concert held on June 11, 2026, including a page that displays a commemorative video of the event.



Desk



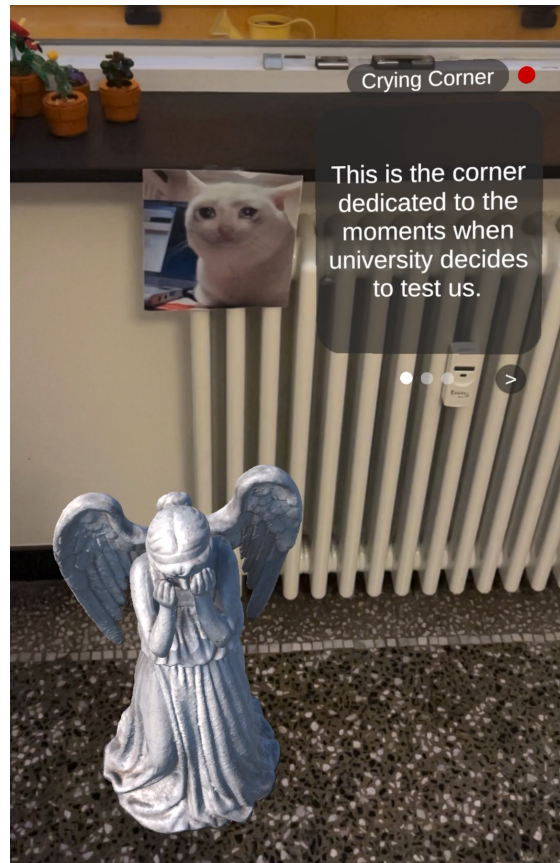
Book



Lewis Hamilton



Tiziano Ferro



Crying Corner

Project Resources

- [Google Drive – Source Code](#)
- [Google Drive – Demo Video](#)